Project Proposal: Accelerating Decision Diffuser for Offline Reinforcement Learning

Group Members: Zihan Xu, Yijie Xu, Yuntong Shi

This project investigates offline reinforcement learning (RL) by framing trajectory generation as a conditional diffusion process, addressing the out-of-distribution action evaluation problem inherent in standard offline RL value-based methods. This approach is compelling because it replaces complex temporal difference learning with a stable generative modeling paradigm, reducing sensitivity to hyperparameters. To provide context, we will examine foundational papers including "Planning with Diffusion for Flexible Behavior Synthesis" (Janner et al., 2022) and "Is Conditional Generative Modeling all you need for Decision-Making?" (Ajay et al., 2023). For our data, we will utilize the open-source D4RL (Datasets for Deep Data-Driven RL) benchmark, specifically focusing on the lightweight `gym-mujoco` subset (e.g., `Hopper-v2` and `HalfCheetah-v2` medium-expert sets) to ensure manageable data processing and training times. Methodologically, rather than building a model from scratch, we propose adapting an existing open-source implementation of the Decision Diffuser. To improve upon the baseline while maintaining a realistic scope, we will implement and analyze the impact of accelerated sampling techniques, specifically DDIM (Denoising Diffusion Implicit Models), to alleviate the slow inference speed typical of diffusion models. Qualitatively, we will evaluate our results by visualizing and comparing the generated state-action trajectories against ground-truth expert trajectories to observe adherence to environmental dynamics. Quantitatively, we will compare the normalized average returns achieved in the simulation environment and measure the computational speedup (inference latency per trajectory) gained by our sampling modification against the standard DDPM baseline.